

Chapter 04 The Laws of Motion

Classical Mechanics

Describes the relationship between the motion of objects and the **forces** acting on them.

Do not apply:

- Very tiny objects(smaller than atomic sizes)
- Objects moving near the speed of light

Forces

Types

- Strong nuclear force
- Electromagnetic force
- Weak nuclear force
- Gravity

All **field forces**, listed in order of decreasing strength. Only **gravity and electromagnetism** is in general physics.

Newton's First Law

An object moves with a velocity that is constant in magnitude and direction, unless acted on by a nonzero net force.

Inertia is the **tendency** of an object to continue in its original motion, relative to mass. Scalar quantity.

Newton's Second Law

The acceleration is directly **proportional to the net force** acting on it and **inversely proportional to its mass**.

$$\vec{a} = \frac{\sum \vec{F}}{m}$$

or

$$\sum \vec{F} = m\vec{a}$$

Gravitational Force

Mutual force of attraction between any two masses.

$$F_g = G \frac{m_1 m_2}{r^2}$$

Weight

The magnitude of the gravitational force acting on an object of mass near the Earth's surface.
(not an inherent property)

Newton's Third Law

If object 1 and object 2 interact, the force exerted by object 1 on object 2 is equal in magnitude but opposite in direction to the force exerted by object 2 on object 1.

$$\vec{F}_{12} = -\vec{F}_{21}$$

External and Internal Forces

External force: Interaction between the object and its environment

Internal force: Within the object itself. **Cannot change the object's velocity.**

Applications of Newton's Laws

Conditions:

- Objects behave as particles
- Masses of strings or ropes are negligible
- Only interested in the forces acting on the object (neglect reaction forces)

Equilibrium

An object at rest or moving with a constant velocity is said to be in equilibrium.
The net force is zero.

$$\sum \vec{F} = 0$$
$$\sum F_x = 0, \sum F_y = 0$$

Forces of Friction

A force resistance to the relative motion, called friction.

For static friction, $f_s = F$

For Kinetic Friction, $f_k = \mu F_n$